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Theorem 0.1. *We can find a non-degenerate curve C on a simple abelian variety A of dimension $n > 2$, where ϕ_C is not generically injective, and there does not exist a non-trivial automorphism $\tau \in \mathrm{Aut}(A)$ which preserves C and acts via a scalar on T_0A.*

1. Idea

For a non-hyperelliptic curve C and a degree 3 line bundle $\mathcal{L}$ with $h^0(\mathcal{L}) = 2$, we know that $\mathcal{L}$ is base-point-free, and it induces a degree 3 map $|\mathcal{L}| : C \longrightarrow \mathbb{P}^1$.

Moreover, the linear system $\left(\mathcal{L}^{\otimes 2}, \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L})\right)$ induces a map $\phi : C \longrightarrow \mathbb{P}^2$ which factor through $|\mathcal{L}|$, so ϕ is not generically injective. If we can find a case $C \subset A$ whose tangent Gauss map is ϕ, then automatically it is not generically injective. For this, we need:

- $\mathcal{L}^{\otimes 2} = \omega_C$;
- $\mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}) \subset \mathrm{H}^0(\omega_C)$ is the cotangent subspace of a subabelian variety.

On the other hand, if we weekened the conditions of non-hyperelliptivity, we can indeed find such a curve: take a hyperelliptic curve C which allows a tangential cover $f : C \longrightarrow E$ at a Weierstrass point p_0. This means that the subspace $\mathrm{Sym}^2 \mathrm{H}^0(\omega_C(-2p_0)) \subset \mathrm{H}^0(\omega_C)$ is the cotangent subspace of a subabelian variety. When $g = 4$, take $\mathcal{L} = \mathcal{O}_C(3p_0)$, we see that

- $\mathcal{L}^{\otimes 2} = \mathcal{O}_C(6p_0) \cong \omega_C$;
- $\mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}) \cong \mathrm{H}^0\left(\mathcal{O}_C(4p_0)\right) = \mathrm{H}^0\left(\omega_C(-2p_0)\right)$ is the cotangent subspace of a subabelian variety.

To solve the non-hyperelliptivity condition, we have to take this hyperelliptic curve as a seed, and then deform near this seed.

2. deformation dimension argument

In this section, we will find a local deformation $(C_t, \mathcal{L}_t)$ of the spin curves where $\mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_t) \subset \mathrm{H}^0(\omega_{C_t})$ still remain as the cotangent subspace of a subabelian variety, while C_t is non-hyperelliptic.

Denote B as the local deformation space of the spin curve $(C_0, \mathcal{L}_0)$, it has dimension

$$\dim B = h^1(\mathcal{T}_C) = 3g - 3 = 9.$$

(1) The set $\Theta_{null} = \left\{(C, \mathcal{L}) \;\middle|\; h^0(\mathcal{L}) \geqslant 2\right\}$ is a divisor;

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(2) Define the rational subspace

$$V_0 := H^1(C_0;\mathbb{Q}) \cap \mathrm{H}^0\left(\omega_C(-2p_0)\right) \subset H^1(C_0;\mathbb{Q}),$$

and take $V_t \subset H^1(C_t;\mathbb{Q})$ as the rational subspace obtained by the deformation via GM connection;

We also define the complex subspace

$$W_0 := \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_\prime) \subset \mathrm{H}^0(\omega_{C_0}),$$

and take $W_t := \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_\sqcup) \subset \mathrm{H}^0(\omega_{C_t})$ as the complex subspace obtained by the deformation via the line bundle.

For the condition that W_t remains as the cotangent subspace of a subabelian variety, we reduce to show

$$\begin{aligned}
&\begin{cases} V_t \text{ remains a Hodge structure} \\ V_{t,\mathbb{C}} \cap \mathrm{H}^0(\omega_C) = W_t \end{cases} \\
\iff\ & V_{t,\mathbb{C}} = W_t \oplus \overline{W}_t \\
\iff\ & W_t \subseteq V_{t,\mathbb{C}} \\
\iff\ & \mathrm{rk}\left[W_t \longrightarrow H^1(C_t;\mathbb{C})/V_{t,\mathbb{C}}\right] = 0
\end{aligned}$$

The last condition cut the deformation space by codimension 6, currently the allowed deformation space has dim $9 - 1 - 6 = 2$.

(3) To avoid the hyperelliptic curve, we have to compute the dimension of local hyperelliptic branch. On a hyperelliptic curve C_t, if $\deg \mathcal{L}_t = 3$ while $h^0(\mathcal{L}_t) = 2$, then we must see that $\mathcal{L}_t \cong \mathcal{O}_{C_t}(3p_0)$ for some Weierestrass point p_0, then we see that

$$W_t = \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_t) = \mathrm{H}^0\left(\omega_{C_t}(-2p_0)\right).$$

The following two lemmas show that the local hyperelliptic branch only has dimension 1:

Lemma 2.1. *For a hyperelliptic curve C, Weierestrass point p_0, $\mathrm{H}^0\left(\omega_{C_t}(-2p_0)\right)$ remains as the cotangent subspace of a subabelian variety, if and only if there exist a tangential cover $f : C \longrightarrow E$ at p_0.*

Lemma 2.2. [1, Proposition 2.5, 3] *Fix an elliptic curve E and integer d, there exists only finitely many tangential covers $f : C \longrightarrow E$ with degree d.*

Therefore, in a 3-dim deformation space, the hyperelliptic locus is at most 2-dimensional, so we can always find a non-hyperelliptic curve satisfying the assumptions.

References

[1] Armando Treibich. Hyperelliptic tangential covers and even elliptic finite-gap potentials, back and forth, 2025.

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